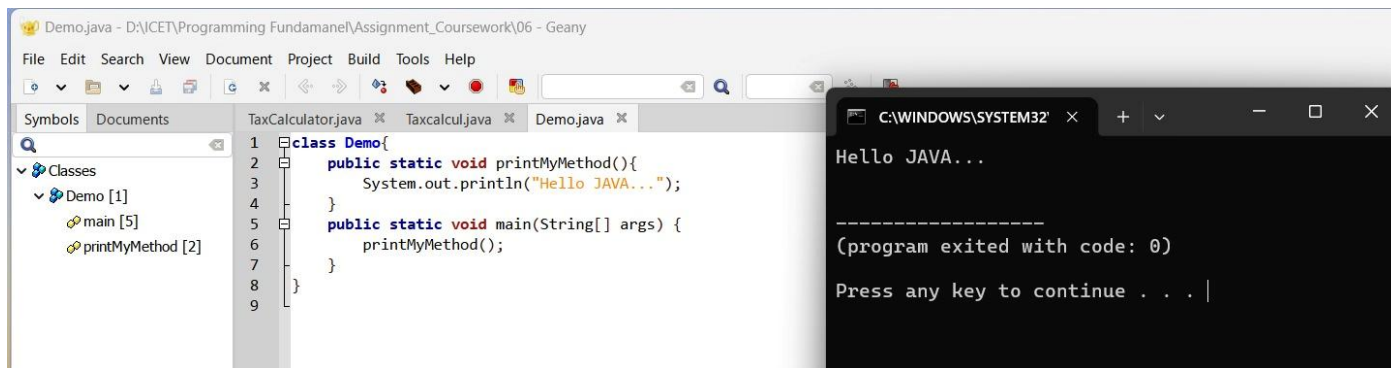


Assignment 06

[PR26121077](#) [Githmi](#) Ththsarani

1.



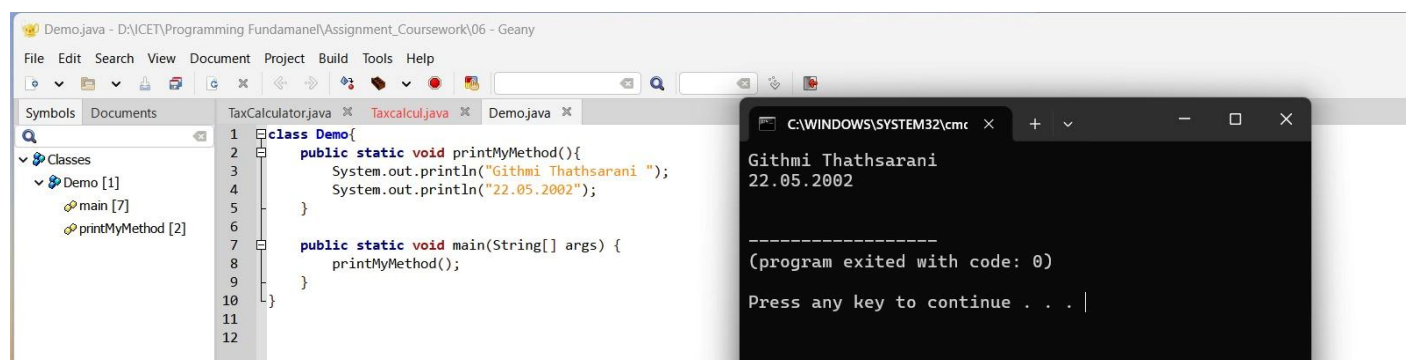
The screenshot shows the Geany IDE with a file named `Demo.java` open. The code defines a `Demo` class with two static methods: `printMyMethod()` and `main()`. The `main` method calls `printMyMethod`. The output window shows the program's execution, printing "Hello JAVA..." and then "(program exited with code: 0)".

```
1 class Demo{
2     public static void printMyMethod(){
3         System.out.println("Hello JAVA...");
4     }
5     public static void main(String[] args) {
6         printMyMethod();
7     }
8 }
9
```

```
Hello JAVA...
-----
(program exited with code: 0)

Press any key to continue . . . |
```

2.



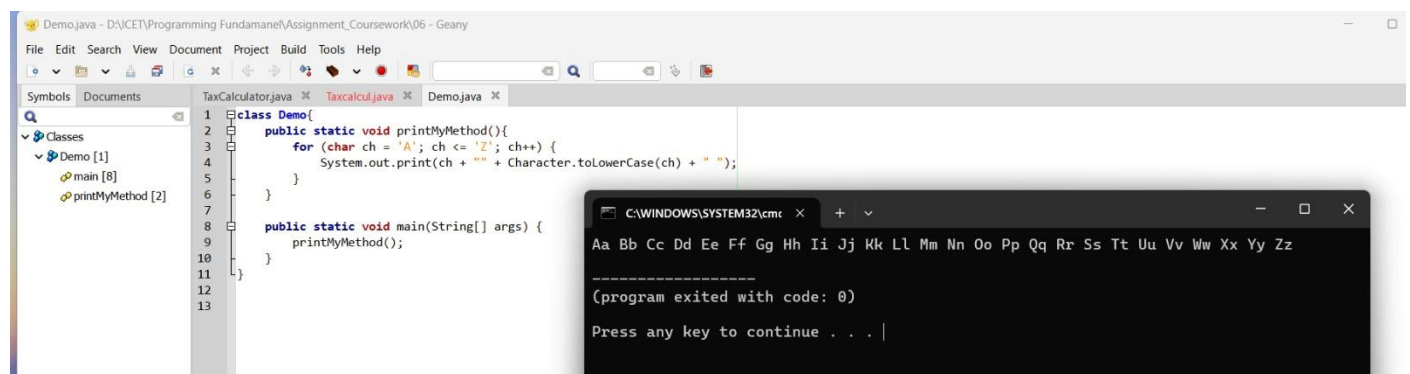
The screenshot shows the Geany IDE with a file named `Demo.java` open. The code defines a `Demo` class with two static methods: `printMyMethod()` and `main()`. The `printMyMethod` method prints the name "Githmi Thathsarani" and the date "22.05.2002". The `main` method calls `printMyMethod`. The output window shows the program's execution, printing the name and date, and then "(program exited with code: 0)".

```
1 class Demo{
2     public static void printMyMethod(){
3         System.out.println("Githmi Thathsarani ");
4         System.out.println("22.05.2002");
5     }
6
7     public static void main(String[] args) {
8         printMyMethod();
9     }
10
11
12
```

```
Githmi Thathsarani
22.05.2002
-----
(program exited with code: 0)

Press any key to continue . . . |
```

3.



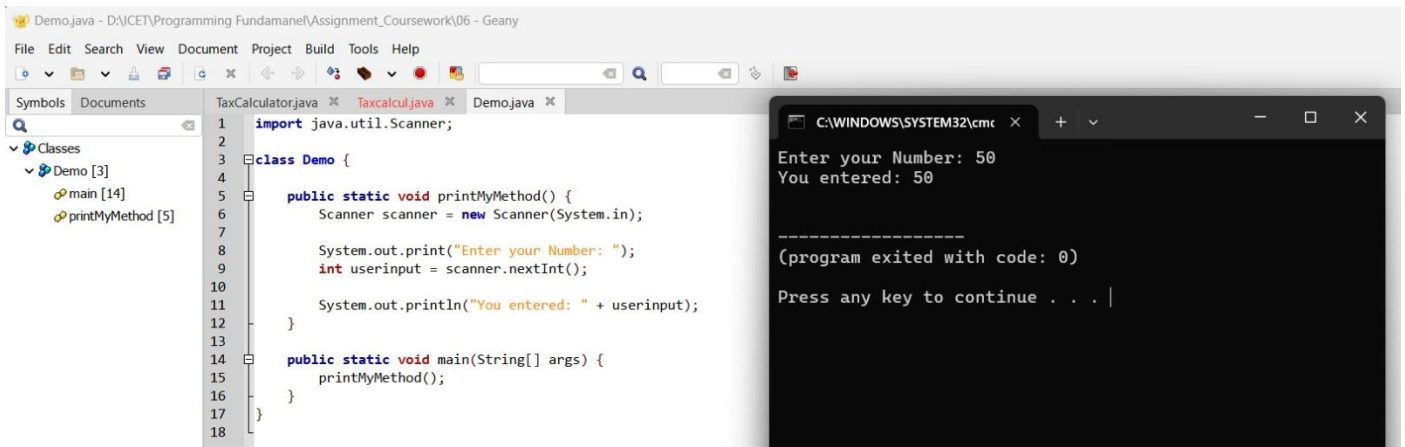
The screenshot shows the Geany IDE with a file named `Demo.java` open. The code defines a `Demo` class with two static methods: `printMyMethod()` and `main()`. The `printMyMethod` method uses a `for` loop to print the lowercase alphabet from 'a' to 'z'. The `main` method calls `printMyMethod`. The output window shows the program's execution, printing the alphabet, and then "(program exited with code: 0)".

```
1 class Demo{
2     public static void printMyMethod(){
3         for (char ch = 'a'; ch <= 'z'; ch++) {
4             System.out.print(ch + " " + Character.toLowerCase(ch) + " ");
5         }
6     }
7
8     public static void main(String[] args) {
9         printMyMethod();
10    }
11
12
13
```

```
Aa Bb Cc Dd Ee Ff Gg Hh Ii Jj Kk Ll Mm Nn Oo Pp Qq Rr Ss Tt Uu Vv Ww Xx Yy Zz
-----
(program exited with code: 0)

Press any key to continue . . . |
```

4.



The screenshot shows an IDE with a file named `Demo.java`. The code is as follows:

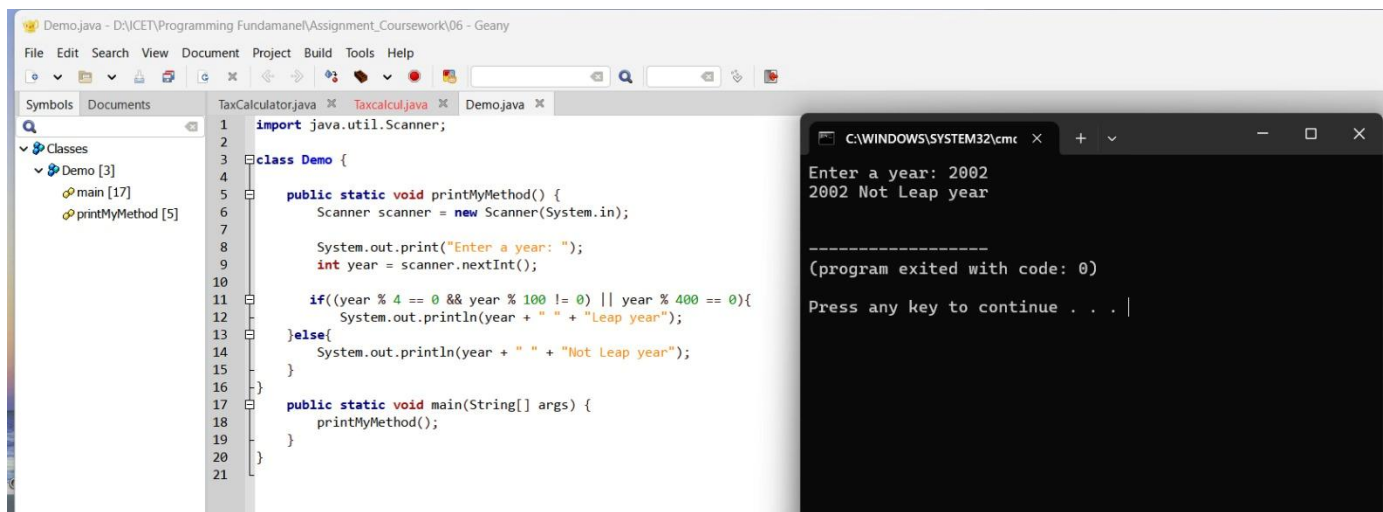
```
1 import java.util.Scanner;
2
3 class Demo {
4
5     public static void printMyMethod() {
6         Scanner scanner = new Scanner(System.in);
7
8         System.out.print("Enter your Number: ");
9         int userInput = scanner.nextInt();
10
11         System.out.println("You entered: " + userInput);
12     }
13
14     public static void main(String[] args) {
15         printMyMethod();
16     }
17 }
18
```

The output window shows the following text:

```
Enter your Number: 50
You entered: 50

-----
(program exited with code: 0)
Press any key to continue . . . |
```

5.



The screenshot shows an IDE with a file named `Demo.java`. The code is as follows:

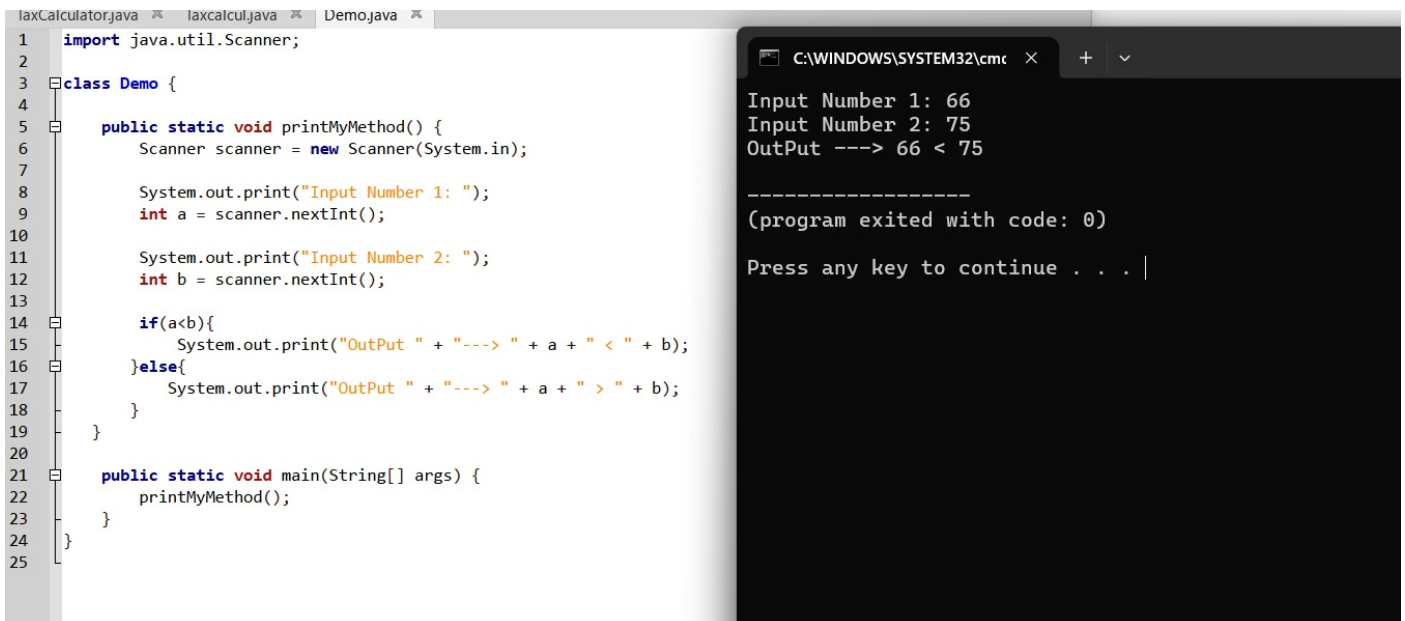
```
1 import java.util.Scanner;
2
3 class Demo {
4
5     public static void printMyMethod() {
6         Scanner scanner = new Scanner(System.in);
7
8         System.out.print("Enter a year: ");
9         int year = scanner.nextInt();
10
11         if((year % 4 == 0 && year % 100 != 0) || year % 400 == 0){
12             System.out.println(year + " " + "Leap year");
13         }else{
14             System.out.println(year + " " + "Not Leap year");
15         }
16     }
17
18     public static void main(String[] args) {
19         printMyMethod();
20     }
21 }
22
```

The output window shows the following text:

```
Enter a year: 2002
2002 Not Leap year

-----
(program exited with code: 0)
Press any key to continue . . . |
```

6.



The screenshot shows an IDE with a file named `Demo.java`. The code is as follows:

```
1 import java.util.Scanner;
2
3 class Demo {
4
5     public static void printMyMethod() {
6         Scanner scanner = new Scanner(System.in);
7
8         System.out.print("Input Number 1: ");
9         int a = scanner.nextInt();
10
11         System.out.print("Input Number 2: ");
12         int b = scanner.nextInt();
13
14         if(a<b){
15             System.out.print("OutPut " + "---> " + a + " < " + b);
16         }else{
17             System.out.print("OutPut " + "---> " + a + " > " + b);
18         }
19     }
20
21     public static void main(String[] args) {
22         printMyMethod();
23     }
24 }
25
```

The output window shows the following text:

```
Input Number 1: 66
Input Number 2: 75
OutPut ---> 66 < 75

-----
(program exited with code: 0)
Press any key to continue . . . |
```

7.

The screenshot shows an IDE window titled "Demo.java - D:\CET\Programming Fundamane\Assignment_Coursework\06 - Geany". The code in the editor is as follows:

```

1  import java.util.Scanner;
2
3  class Demo {
4
5      public static void printMyMethod() {
6          Scanner scanner = new Scanner(System.in);
7
8          System.out.print("Input first word: ");
9          String word1 = scanner.nextLine();
10
11         System.out.print("Input second word: ");
12         String word2 = scanner.nextLine();
13
14         if(word1.charAt(0) == word2.charAt(0)){
15             System.out.println("True");
16         } else {
17             System.out.println("False");
18         }
19     }
20
21     public static void main(String[] args) {
22         printMyMethod();
23     }
24 }
25
26

```

The terminal output on the right shows the program's execution:

```

C:\WINDOWS\SYSTEM32\cmd
Input first word: abcd
Input second word: efgh
False

-----
(program exited with code: 0)
Press any key to continue . . . |

```

8.

The screenshot shows an IDE window titled "Demo.java - D:\CET\Programming Fundamane\Assignment_Coursework\06 - Geany". The code in the editor is as follows:

```

1  import java.util.Scanner;
2
3  class Demo {
4
5      public static double roundToTwoDecimalPlaces(double number) {
6          return Math.round(number * 100.0) / 100.0;
7      }
8
9      public static void main(String[] args) {
10         Scanner scanner = new Scanner(System.in);
11
12         System.out.print("Enter a decimal number: ");
13         double input = scanner.nextDouble();
14
15         double result = roundToTwoDecimalPlaces(input);
16
17         System.out.println("Original number: " + input);
18         System.out.println("Rounded to 2 decimal places: " + result);
19
20         scanner.close();
21     }
22 }
23
24

```

The terminal output on the right shows the program's execution:

```

C:\WINDOWS\SYSTEM32\cmd
Enter a decimal number: 8765.3548
Original number: 8765.3548
Rounded to 2 decimal places: 8765.35

-----
(program exited with code: 0)
Press any key to continue . . . |

```

9.

```

1 import java.util.Scanner;
2
3 class Demo {
4
5     public static void printMyMethod(){
6         Scanner scanner = new Scanner(System.in);
7
8         System.out.print("Enter Month Salary: ");
9         double salary = scanner.nextDouble();
10
11         double anualinterest = 20;
12
13         double monthlyInterst = (salary * anualinterest / 100)/12.0;
14
15         System.out.print("Monthly Interst " + monthlyInterst);
16
17
18         scanner.close();
19     }
20
21
22     public static void main(String[] args) {
23         printMyMethod();
24     }
25 }
26
27

```

Enter Month Salary: 75000.00
Monthly Interst 1250.0

(program exited with code: 0)
Press any key to continue . . . |

10.

```

1 import java.util.Scanner;
2
3 class Demo {
4
5     public static double celsiusToFahrenheit(double celsius){
6         return celsius * (9.0 / 5.0) + 32;
7     }
8
9     public static double celsiusToKelvin(double celsius){
10        return celsius + 273.15;
11    }
12
13    public static void convertAndDisplay(double celsius){
14        double fahrenheit = celsiusToFahrenheit(celsius);
15        double kelvin = celsiusToKelvin(celsius);
16
17        System.out.println("\n--- Conversion Results ---");
18        System.out.println(celsius+"°C" + fahrenheit + "°F");
19        System.out.println(celsius+"°C" + kelvin + "°K");
20    }
21
22    public static void main (String[] args) {
23        Scanner scanner = new Scanner(System.in);
24
25        System.out.print("Enter temperature in Celsius: ");
26        double celsius = scanner.nextDouble();
27
28        convertAndDisplay(celsius);
29
30        scanner.close();
31    }
32 }
33
34

```

Enter temperature in Celsius: 35
--- Conversion Results ---
35.0°C95.0°F
35.0°C308.15°K

(program exited with code: 0)
Press any key to continue . . . |

11.

The screenshot shows an IDE window titled "Demo.java - D:\JCET\Programming Fundamane\Assignment_Coursework\06 - Geany". The code in the editor is as follows:

```

1 import java.util.Scanner;
2
3 class Demo {
4
5     public static double celsiusToFahrenheit(double celsius){
6         return celsius * (9.0 / 5.0) + 32;
7     }
8
9     public static double celsiusToKelvin(double celsius){
10        return celsius + 273.15;
11    }
12
13    public static void convertAndDisplay(double celsius){
14        double fahrenheit = celsiusToFahrenheit(celsius);
15        double kelvin = celsiusToKelvin(celsius);
16
17        System.out.println("\n--- Conversion Results ---");
18        System.out.println(celsius+"°C" + fahrenheit + "°F");
19        System.out.println(celsius+"°C" + kelvin + "°K");
20    }
21
22    public static void main (String[] args) {
23        Scanner scanner = new Scanner(System.in);
24
25        System.out.print("Enter temperature in Celsius: ");
26        double celsius = scanner.nextDouble();
27
28        convertAndDisplay(celsius);
29
30        scanner.close();
31    }
32 }
33
34

```

The terminal window on the right shows the program's execution:

```

C:\WINDOWS\SYSTEM32\cmd
Enter temperature in Celsius: 35

--- Conversion Results ---
35.0°C95.0°F
35.0°C308.15°K

(program exited with code: 0)
Press any key to continue . . .

```

12.

The screenshot shows an IDE window titled "Demo.java - D:\JCET\Programming Fundamane\Assignment_Coursework\06 - Geany". The code in the editor is as follows:

```

1 import java.util.Scanner;
2
3 public class Demo {
4
5     public static void areaofasector(double angle, double radius) {
6         double angleRad = Math.toRadians(angle);
7         double area = (angle / 360) * (22.0 / 7.0) * radius * radius;
8         double perimeter = 2 * radius + (angle / 360) * 2 * Math.PI * radius;
9         System.out.println("Area: " + area + ", Perimeter: " + perimeter);
10    }
11
12    public static void main(String args[]) {
13        Scanner scanner = new Scanner(System.in);
14
15        // Get user input
16        System.out.print("Enter angle in degrees: ");
17        double angle = scanner.nextDouble();
18
19        System.out.print("Enter radius: ");
20        double radius = scanner.nextDouble();
21
22        // Call method with proper arguments
23        areaofasector(angle, radius);
24
25        scanner.close();
26    }
27 }
28

```

The terminal window on the right shows the program's execution:

```

C:\WINDOWS\SYSTEM32\cmd
Enter angle in degrees: 89
Enter radius: 75
Area: 4370.535714285715, Perimeter: 266.5007275706215

(program exited with code: 0)
Press any key to continue . . .

```

13.

```

1 import java.util.Scanner;
2 class Demo {
3     public static void findQuadratic() {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter root a: ");
6         int a = sc.nextInt();
7         System.out.print("Enter root b: ");
8         int b = sc.nextInt();
9         System.out.println("x2 - (" + (a+b) + ")x + (" + (a*b) + ") = 0");
10    }
11    public static void main(String[] args) {
12        findQuadratic();
13    }
14 }
15
16

```

```

C:\WINDOWS\SYSTEM32\cmd
Enter root a: 10
Enter root b: 5
x2 - (15)x + (50) = 0

-----
(program exited with code: 0)
Press any key to continue . . . |

```

14.

15.

```

1 import java.util.Scanner;
2 class Demo {
3     // Method to swap two numbers without using third variable
4     public void swapNumbers(int a, int b) {
5         System.out.println("Before swapping:");
6         System.out.println("First number: " + a);
7         System.out.println("Second number: " + b);
8
9         // Swapping without using third variable
10        a = a + b;
11        b = a - b;
12        a = a - b;
13
14        System.out.println("\nAfter swapping:");
15        System.out.println("First number: " + a);
16        System.out.println("Second number: " + b);
17    }
18
19    public static void main(String[] args) {
20        Scanner scanner = new Scanner(System.in);
21        Demo demo = new Demo();
22
23        System.out.print("Enter first number: ");
24        int num1 = scanner.nextInt();
25
26        System.out.print("Enter second number: ");
27        int num2 = scanner.nextInt();
28
29        // Call the swap method
30        demo.swapNumbers(num1, num2);
31
32        scanner.close();
33    }
34 }
35
36

```

```

C:\WINDOWS\SYSTEM32\cmd
Enter first number: 80
Enter second number: 90
Before swapping:
First number: 80
Second number: 90

After swapping:
First number: 90
Second number: 80

-----
(program exited with code: 0)
Press any key to continue . . . |

```

16.

```

1 class Demo {
2     public static boolean isPerfect(int n) {
3         int sum = 0;
4         for (int i = 1; i <= n/2; i++) {
5             if (n % i == 0) sum += i;
6         }
7         return sum == n;
8     }
9     public static void main(String[] args) {
10        System.out.println(isPerfect(6)); // true
11        System.out.println(isPerfect(28)); // true
12    }
13 }
14

```

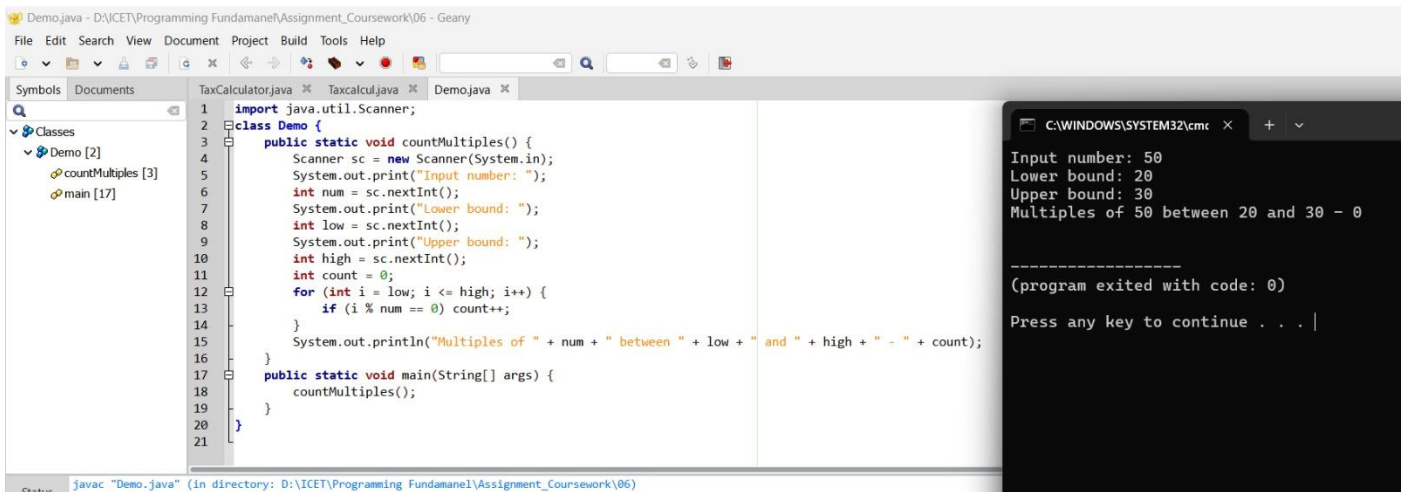
```

C:\WINDOWS\SYSTEM32\cmd
true
true

-----
(program exited with code: 0)
Press any key to continue . . . |

```

17.



```

1 import java.util.Scanner;
2 class Demo {
3     public static void countMultiples() {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Input number: ");
6         int num = sc.nextInt();
7         System.out.print("Lower bound: ");
8         int low = sc.nextInt();
9         System.out.print("Upper bound: ");
10        int high = sc.nextInt();
11        int count = 0;
12        for (int i = low; i <= high; i++) {
13            if (i % num == 0) count++;
14        }
15        System.out.println("Multiples of " + num + " between " + low + " and " + high + " - " + count);
16    }
17    public static void main(String[] args) {
18        countMultiples();
19    }
20 }
21

```

javac "Demo.java" (in directory: D:\CET\Programming Fundamentals\Assignment_Coursework\06)

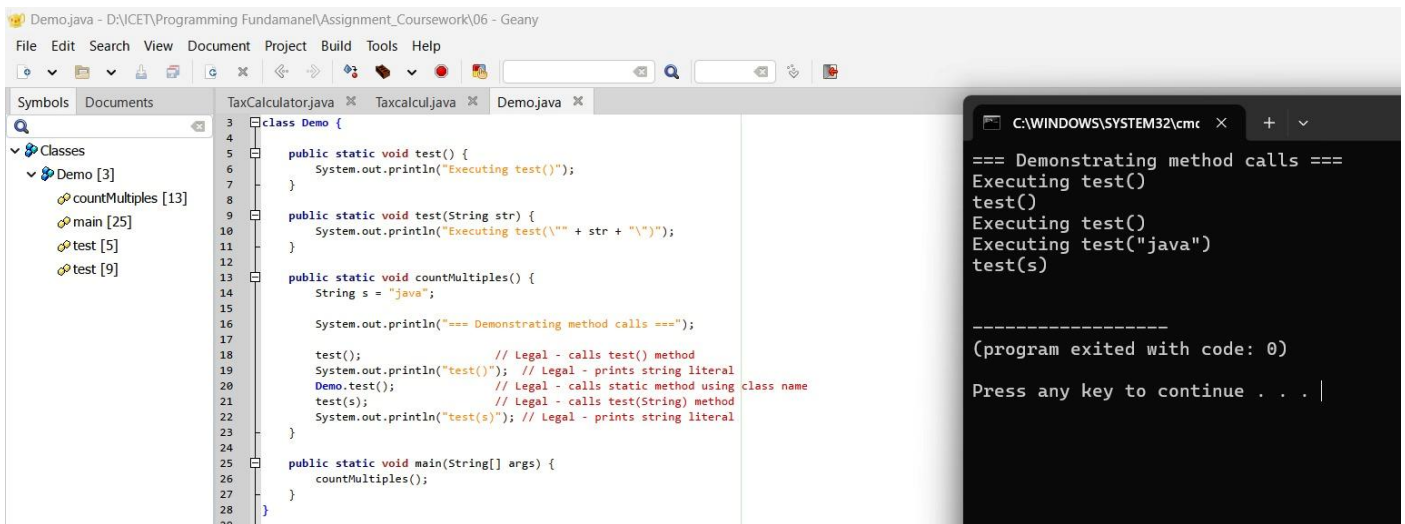
```

C:\WINDOWS\SYSTEM32\cmd
Input number: 50
Lower bound: 20
Upper bound: 30
Multiples of 50 between 20 and 30 - 0

-----
(program exited with code: 0)
Press any key to continue . . . |

```

18.



```

3 class Demo {
4     public static void test() {
5         System.out.println("Executing test()");
6     }
7
8     public static void test(String str) {
9         System.out.println("Executing test(\"" + str + "\")");
10    }
11
12    public static void countMultiples() {
13        String s = "java";
14
15        System.out.println("=== Demonstrating method calls ===");
16
17        test(); // Legal - calls test() method
18        System.out.println("test()"); // Legal - prints string literal
19        Demo.test(); // Legal - calls static method using class name
20        test(s); // Legal - calls test(String) method
21        System.out.println("test(s)"); // Legal - prints string literal
22    }
23
24    public static void main(String[] args) {
25        countMultiples();
26    }
27
28
29

```

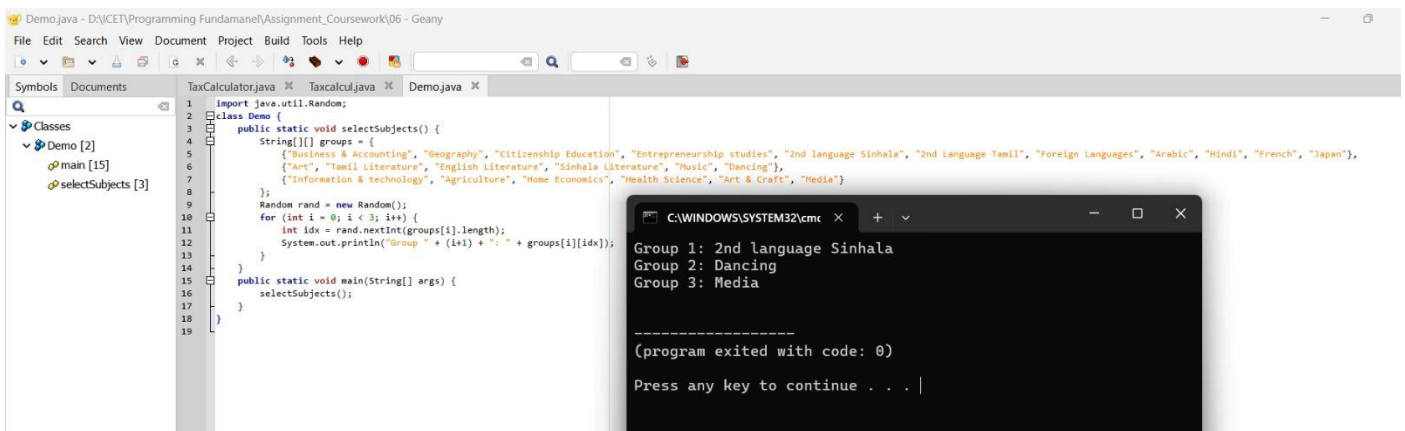
```

C:\WINDOWS\SYSTEM32\cmd
=== Demonstrating method calls ===
Executing test()
test()
Executing test()
Executing test("java")
test(s)

-----
(program exited with code: 0)
Press any key to continue . . . |

```

19.



```

1 import java.util.Random;
2 class Demo {
3     public static void selectSubjects() {
4         String[] groups = {
5             "Business & Accounting", "Geography", "Citizenship Education", "Entrepreneurship studies", "2nd language Sinhala", "2nd Language Tamil", "Foreign Languages", "Arabic", "Hindi", "French", "Japan",
6             "Art", "Tamil Literature", "English Literature", "Sinhala Literature", "Music", "Dancing",
7             "Information & technology", "Agriculture", "Home Economics", "Health Science", "Art & Craft", "Media"
8         };
9         Random rand = new Random();
10        for (int i = 0; i < 3; i++) {
11            int idx = rand.nextInt(groups[i].length);
12            System.out.println("Group " + (i+1) + ": " + groups[i][idx]);
13        }
14    }
15    public static void main(String[] args) {
16        selectSubjects();
17    }
18
19

```

```

C:\WINDOWS\SYSTEM32\cmd
Group 1: 2nd language Sinhala
Group 2: Dancing
Group 3: Media

-----
(program exited with code: 0)
Press any key to continue . . . |

```

20.

Line 2 (testMethod2();) — error: non-static method called from static context (unless testMethod2 is static, but here it's not).

Line 5 (Demo.testMethod2();) — error: cannot call non-static method statically.

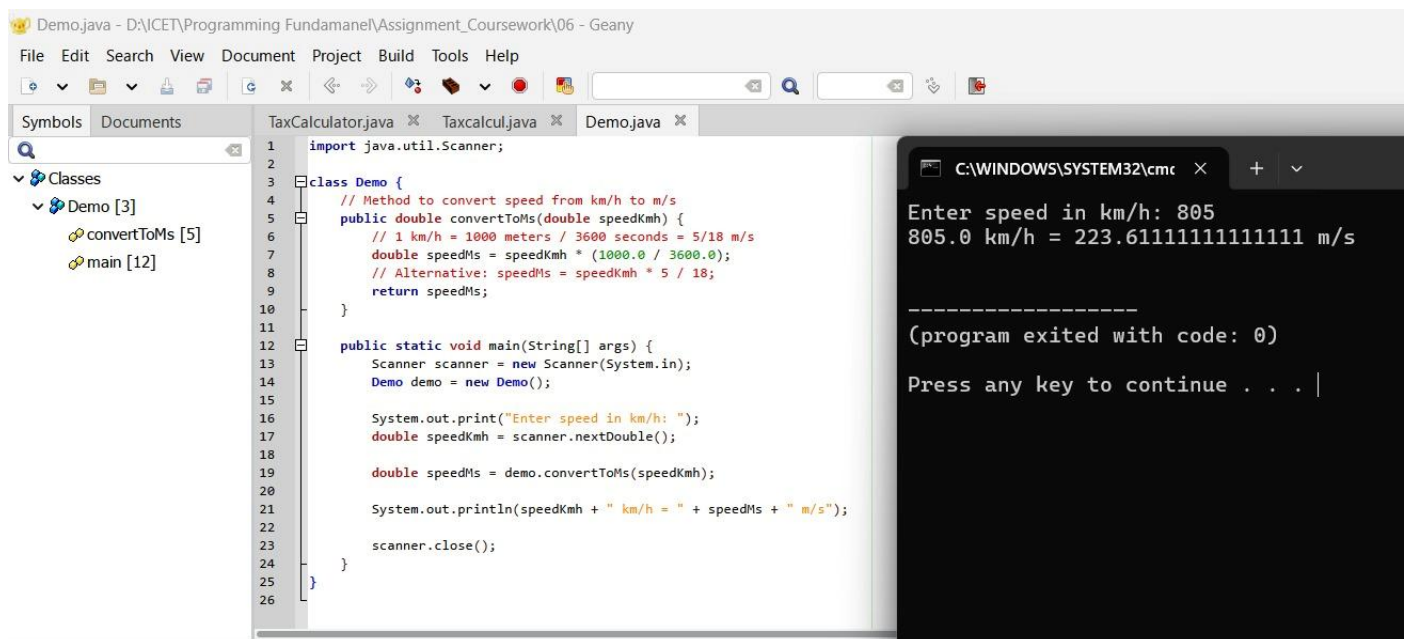
Line 6 (Demo d = new Demo();) — legal if Demo is not inner class of Test with issues; but here Demo is inner class of Test, so must be Test.Demo d = new Test().new Demo();.

21.

Legal: a, b (if in class), f, g (if semicolon removed), k, l.

Illegal: c (void after static), d (no name), e (no name), h (no type for x), i (literal as param), j (void but returns 0).

22.



The screenshot shows the Geany IDE with a project named 'Demo.java'. The main editor displays the following Java code:

```
1 import java.util.Scanner;
2
3 class Demo {
4     // Method to convert speed from km/h to m/s
5     public double convertToMs(double speedKmh) {
6         // 1 km/h = 1000 meters / 3600 seconds = 5/18 m/s
7         double speedMs = speedKmh * (1000.0 / 3600.0);
8         // Alternative: speedMs = speedKmh * 5 / 18;
9         return speedMs;
10    }
11
12    public static void main(String[] args) {
13        Scanner scanner = new Scanner(System.in);
14        Demo demo = new Demo();
15
16        System.out.print("Enter speed in km/h: ");
17        double speedKmh = scanner.nextDouble();
18
19        double speedMs = demo.convertToMs(speedKmh);
20
21        System.out.println(speedKmh + " km/h = " + speedMs + " m/s");
22
23        scanner.close();
24    }
25 }
26
```

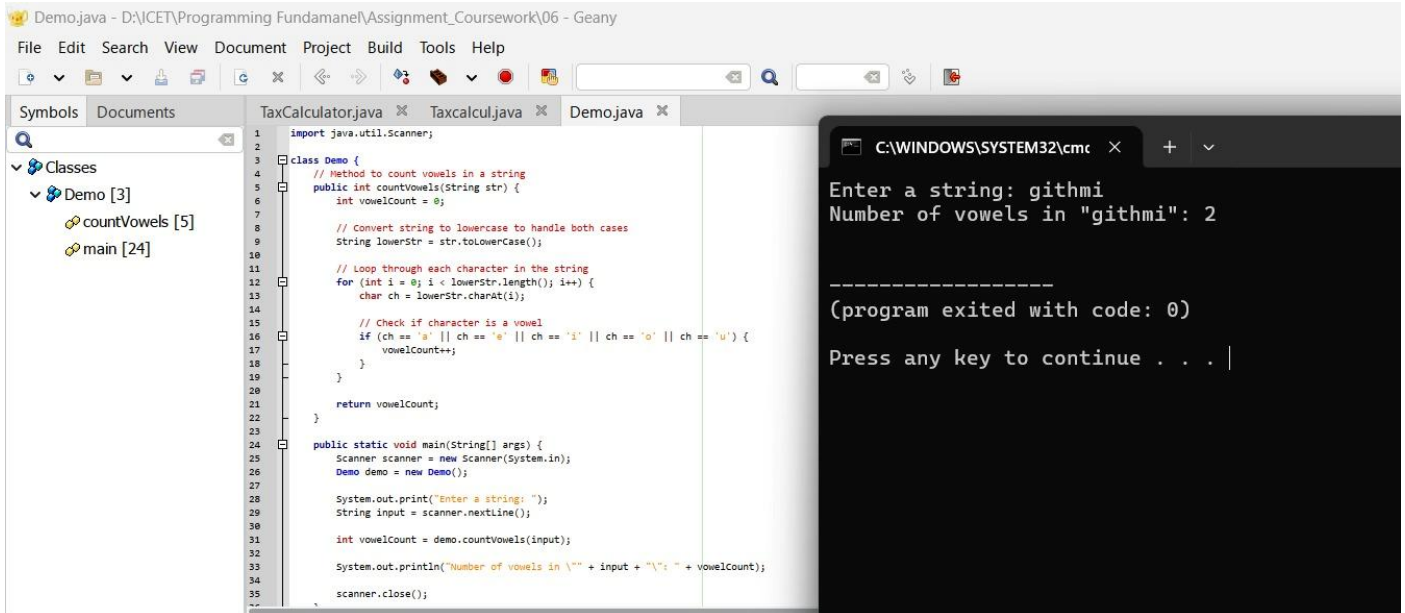
The left sidebar shows the 'Classes' pane with 'Demo [3]' expanded, showing 'convertToMs [5]' and 'main [12]'. The right pane shows the terminal output:

```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter speed in km/h: 805
805.0 km/h = 223.61111111111111 m/s

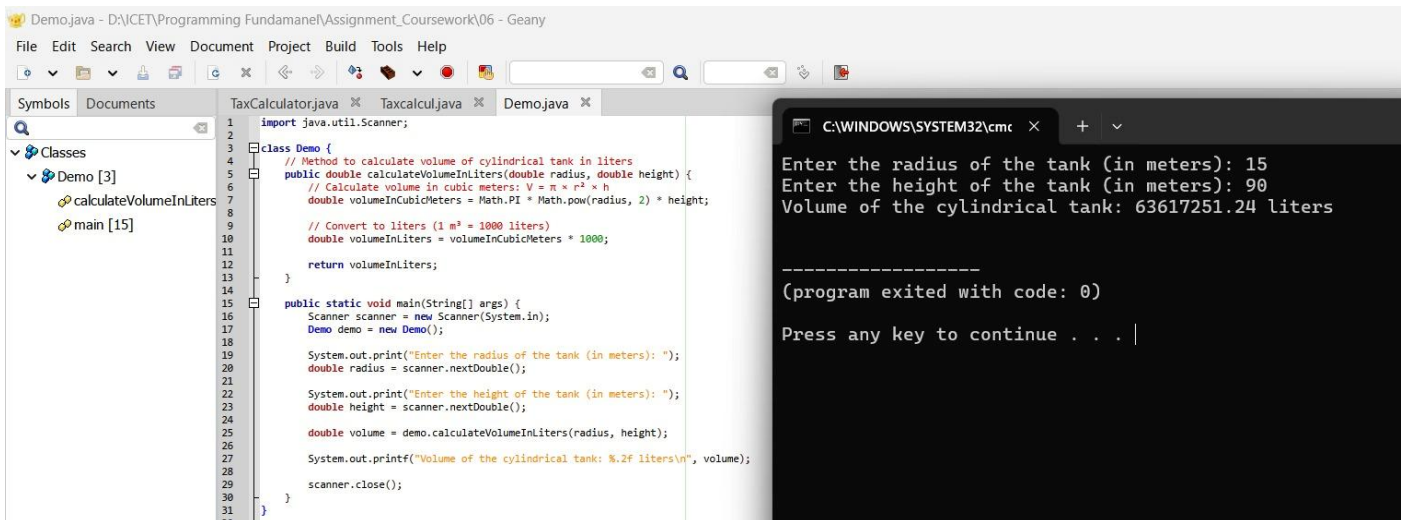
-----
(program exited with code: 0)

Press any key to continue . . . |
```

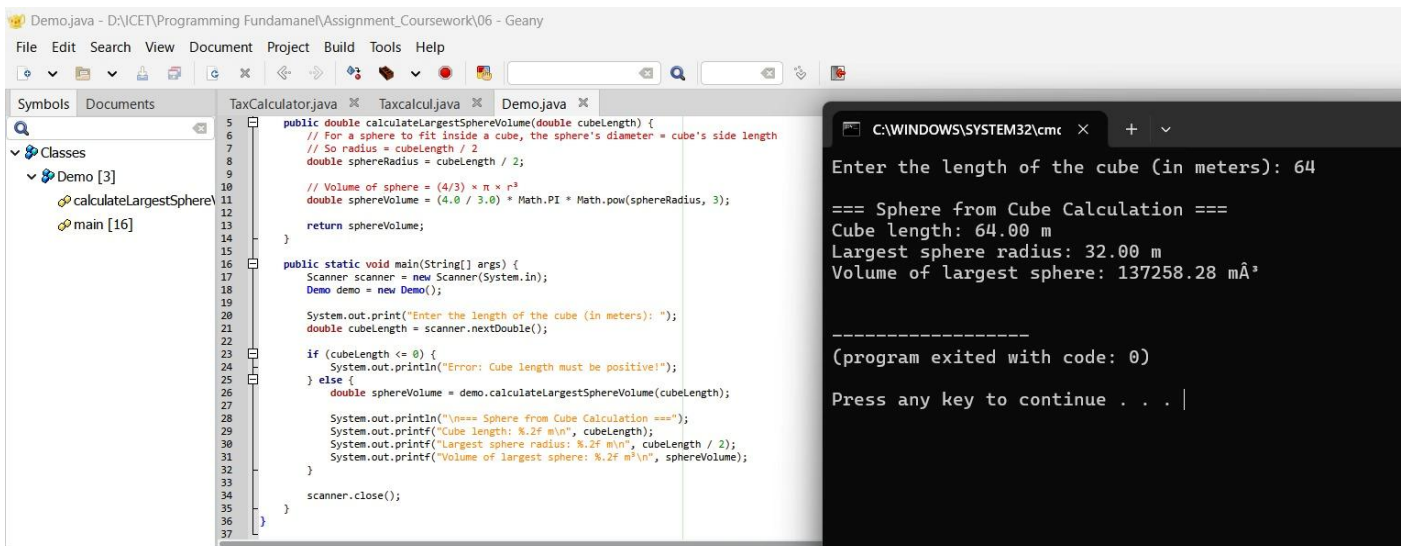
23.



24.



25.



26.

Line 1: x : 100 - Prints the original value of x in main

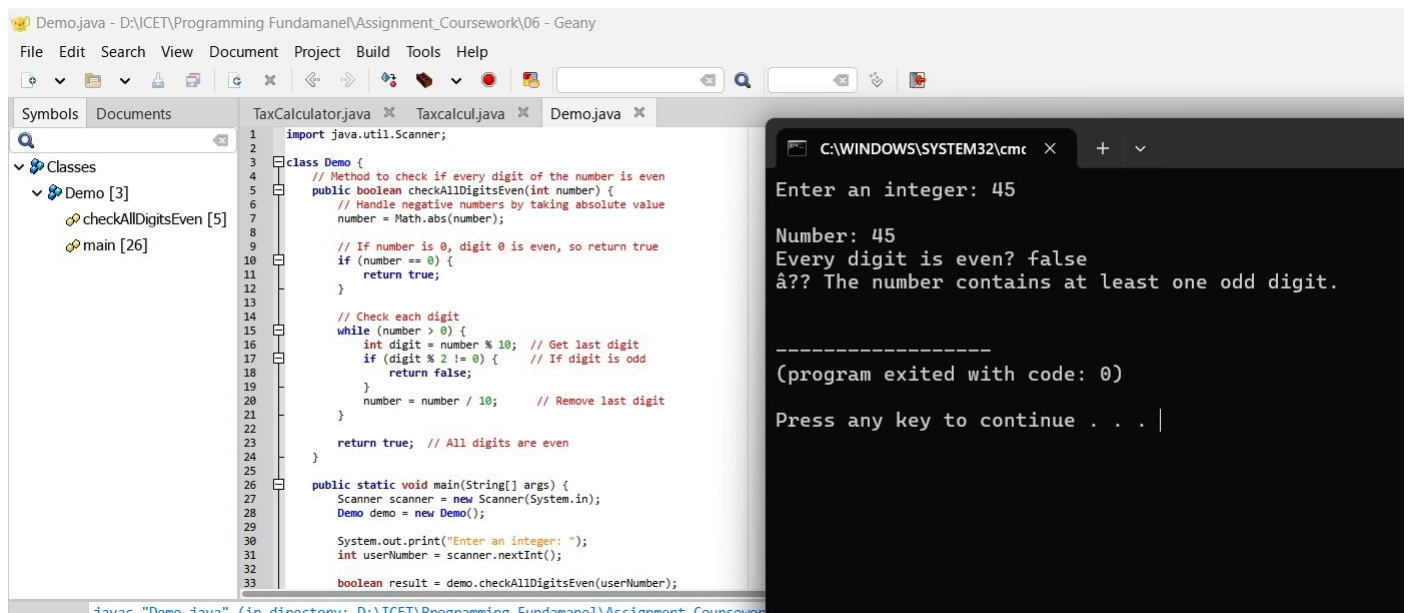
Line 2: x : 101 - Inside increment() method, x is incremented from 100 to 101

Line 3: x : 100 - Back in main, original x remains 100 (pass by value - changes inside method don't affect original)

Line 4: x : 101 - Inside increment() again, x is incremented from 100 to 101

Line 5: x : 101 - This time the returned value (101) is assigned to x in main, so x becomes 101

27.



The screenshot shows an IDE with a Java file named 'Demo.java'. The code defines a class 'Demo' with a method 'checkAllDigitsEven' that checks if all digits of a number are even. The 'main' method uses a 'Scanner' to take user input and calls the 'checkAllDigitsEven' method. The output window shows the program running with input '45', outputting 'Number: 45', 'Every digit is even? false', and 'â?? The number contains at least one odd digit.'.

```
1 import java.util.Scanner;
2
3 class Demo {
4     // Method to check if every digit of the number is even
5     public boolean checkAllDigitsEven(int number) {
6         // Handle negative numbers by taking absolute value
7         number = Math.abs(number);
8
9         // If number is 0, digit 0 is even, so return true
10        if (number == 0) {
11            return true;
12        }
13
14        // Check each digit
15        while (number > 0) {
16            int digit = number % 10; // Get last digit
17            if (digit % 2 != 0) { // If digit is odd
18                return false;
19            }
20            number = number / 10; // Remove last digit
21        }
22
23        return true; // All digits are even
24    }
25
26    public static void main(String[] args) {
27        Scanner scanner = new Scanner(System.in);
28        Demo demo = new Demo();
29
30        System.out.print("Enter an integer: ");
31        int userNumber = scanner.nextInt();
32
33        boolean result = demo.checkAllDigitsEven(userNumber);
```

Enter an integer: 45
Number: 45
Every digit is even? false
â?? The number contains at least one odd digit.

(program exited with code: 0)
Press any key to continue . . . |

28.

B, C, D, F, G can all be inserted legally.

29.

```

1 import java.util.Scanner;
2
3 class Demo {
4     // Method to get number of days in a month
5     public int getDaysInMonth(String monthName) {
6         // Convert to lowercase for case-insensitive comparison
7         String month = monthName.toLowerCase();
8
9         // Months with 31 days
10        if (month.equals("january") || month.equals("march") || month.equals("may") ||
11            month.equals("july") || month.equals("august") || month.equals("october") ||
12            month.equals("december")) {
13            return 31;
14        }
15
16        // Months with 30 days
17        else if (month.equals("april") || month.equals("june") || month.equals("september") ||
18            month.equals("november")) {
19            return 30;
20        }
21
22        // February
23        else if (month.equals("february")) {
24            return 28; // Non-leap year
25        }
26
27        // Invalid month name
28        else {
29            System.out.println("month name incorrect! ");
30            return -1;
31        }
32    }
33 }

```

C:\WINDOWS\SYSTEM32\cmd.exe

```

Enter month name: August
August has 31 days.

-----
(program exited with code: 0)

Press any key to continue . . . |

```

30.

```

1 import java.util.Scanner;
2
3 class Demo {
4     // Method to separate even and odd digits
5     public void separateEvenOddDigits(int number) {
6         // Handle negative numbers by converting to positive
7         if (number < 0) {
8             System.out.println("Please enter a positive integer!");
9             return;
10        }
11
12        // Convert number to string to preserve digit order
13        String numStr = Integer.toString(number);
14
15        // StringBuilder for efficient string concatenation
16        StringBuilder oddDigits = new StringBuilder();
17        StringBuilder evenDigits = new StringBuilder();
18
19        // Check each digit
20        for (int i = 0; i < numStr.length(); i++) {
21            char digitChar = numStr.charAt(i);
22            int digit = Character.getNumericValue(digitChar);
23
24            if (digit % 2 == 0) {
25                evenDigits.append(digit).append(" ");
26            } else {
27                oddDigits.append(digit).append(" ");
28            }
29        }
30
31        // Print results
32        System.out.println("Input: " + number);
33        System.out.print("Odd digits: " + oddDigits.toString().trim());

```

C:\WINDOWS\SYSTEM32\cmd.exe

```

Enter a positive integer: 62
Input: 62
Odd digits:
Even digits: 6 2

-----
(program exited with code: 0)

Press any key to continue . . . |

```

31.

```
1 import java.util.Random;
2
3 class Demo {
4     // Method to simulate rolling two dice until they match
5     public void rollUntilMatch() {
6         Random random = new Random();
7         int rollCount = 0;
8         int die1, die2;
9
10        System.out.println("Rolling two dice until they match...\n");
11
12        do {
13            // Roll both dice (1 to 6)
14            die1 = random.nextInt(6) + 1;
15            die2 = random.nextInt(6) + 1;
16            rollCount++;
17
18            // Print the result of each roll
19            System.out.println(die1 + " " + die2);
20
21        } while (die1 != die2);
22
23        System.out.println("\nDice roll - " + rollCount + " times");
24    }
25
26    public static void main(String[] args) {
27        Demo demo = new Demo();
28        demo.rollUntilMatch();
29    }
30
31 }
```

```
Rolling two dice until they match...
5 3
6 6
Dice roll - 2 times
(program exited with code: 0)
Press any key to continue . . . |
```

32.

A, B, C, G

33.

C, D, E, F, H, K, O

34.

```
3 public class Demo {
4     // Method to calculate factorial
5     public static long factorial(int num) {
6         long fact = 1;
7         for (int i = 1; i <= num; i++) {
8             fact *= i;
9         }
10        return fact;
11    }
12
13    // Method to calculate number of groups
14    public static long calculateGroups(int n, int r) {
15        if (r > n) {
16            System.out.println("Group size cannot be greater than number of children.");
17            return 0;
18        }
19        long numerator = factorial(n);
20        long denominator = factorial(n - r) * factorial(r);
21        return numerator / denominator;
22    }
23
24    public static void main(String[] args) {
25        Scanner input = new Scanner(System.in);
26
27        System.out.print("Number of children (n): ");
28        int n = input.nextInt();
29
30        System.out.print("Number of group members (r): ");
31        int r = input.nextInt();
32
33        long groups = calculateGroups(n, r);
34        System.out.println("Number of groups = " + groups);
35    }
36 }
```

```
Number of children (n): 50
Number of group members (r): 73
Group size cannot be greater than number of children.
Number of groups = 0
(program exited with code: 0)
Press any key to continue . . . |
```

35.

The screenshot shows the Geany IDE with a file named `Demo.java` open. The file is located at `D:\ICET\Programming Fundamanel\Assignment_Coursework\06`. The code in `Demo.java` defines a `public class Demo` with three methods: `gcd` (using Euclidean algorithm), `lcm` (product of two numbers divided by their GCD), and `smallestDivisibleByRange` (finding the smallest number divisible by all numbers in a range). The `main` method calls `smallestDivisibleByRange(20)` and prints the result.

```

1 public class Demo {
2
3     // Method to find GCD using Euclidean algorithm
4     public static long gcd(long a, long b) {
5         while (b != 0) {
6             long temp = b;
7             b = a % b;
8             a = temp;
9         }
10        return a;
11    }
12
13    // Method to find LCM of two numbers
14    public static long lcm(long a, long b) {
15        return (a * b) / gcd(a, b);
16    }
17
18    // Method to find smallest number divisible by 1 to n
19    public static long smallestDivisibleByRange(int n) {
20        long result = 1;
21        for (int i = 2; i <= n; i++) {
22            result = lcm(result, i);
23        }
24        return result;
25    }
26
27    public static void main(String[] args) {
28        long answer = smallestDivisibleByRange(20);
29        System.out.println("Smallest number divisible by 1 to 20: " + answer);
30    }
31 }
32

```

The terminal window shows the output of the program:

```

C:\WINDOWS\SYSTEM32\cmd.exe
Smallest number divisible by 1 to 20: 232792560

-----
(program exited with code: 0)
Press any key to continue . . .

```

36.

The screenshot shows the Geany IDE with a file named `Demo.java` open. The file is located at `D:\ICET\Programming Fundamanel\Assignment_Coursework\06`. The code in `Demo.java` defines a `public class Demo` with a method `findQuadrantAndDistance` that takes `double x` and `double y` as input and returns the quadrant and distance from the origin. The `main` method uses a `Scanner` to take user input for `x` and `y` and calls `findQuadrantAndDistance`.

```

1 import java.util.Scanner;
2
3 public class Demo {
4
5     public static void findQuadrantAndDistance(double x, double y) {
6         // Calculate distance from origin
7         double distance = Math.sqrt(x * x + y * y);
8
9         // Determine location
10        if (x == 0 && y == 0) {
11            System.out.println("Origin");
12        } else if (x == 0) {
13            System.out.println("On Y-axis");
14        } else if (y == 0) {
15            System.out.println("On X-axis");
16        } else if (x > 0 && y > 0) {
17            System.out.println("Quadrant I");
18        } else if (x < 0 && y > 0) {
19            System.out.println("Quadrant II");
20        } else if (x < 0 && y < 0) {
21            System.out.println("Quadrant III");
22        } else if (x > 0 && y < 0) {
23            System.out.println("Quadrant IV");
24        }
25
26        System.out.printf("Distance from (0,0): %.2f\n", distance);
27    }
28
29    public static void main(String[] args) {
30        Scanner input = new Scanner(System.in);
31
32        System.out.print("Enter X coordinate: ");
33        double x = input.nextDouble();
34    }
35 }
36

```

The terminal window shows the output of the program:

```

C:\WINDOWS\SYSTEM32\cmd.exe
Enter X coordinate: 50
Enter Y coordinate: 40
Quadrant I
Distance from (0,0): 64.03

-----
(program exited with code: 0)
Press any key to continue . . .

```

37.

```

1 import java.util.Scanner;
2
3 public class Demo {
4
5     // Method to check triangle validity and return perimeter
6     public static double checkTriangleAndGetPerimeter(double a, double b, double c) {
7         // Triangle validity conditions
8         if (a + b > c && a + c > b && b + c > a) {
9             // Valid triangle: calculate and return perimeter
10            double perimeter = a + b + c;
11            System.out.println("Valid triangle");
12            return perimeter;
13        } else {
14            // Invalid triangle
15            System.out.println("Invalid triangle");
16            return -1; // Return -1 to indicate invalid triangle
17        }
18    }
19
20    public static void main(String[] args) {
21        Scanner input = new Scanner(System.in);
22
23        System.out.print("Enter first side: ");
24        double side1 = input.nextDouble();
25
26        System.out.print("Enter second side: ");
27        double side2 = input.nextDouble();
28
29        System.out.print("Enter third side: ");
30        double side3 = input.nextDouble();
31
32        double perimeter = checkTriangleAndGetPerimeter(side1, side2, side3);
33    }
34 }

```

```

C:\WINDOWS\SYSTEM32\cmd
Enter first side: 23
Enter second side: 21
Enter third side: 20
Valid triangle
Perimeter: 64.00

-----
(program exited with code: 0)

Press any key to continue . . .

```

38.

```

1 import java.util.Scanner;
2
3 public class Demo {
4
5     // Method to check triangle validity and return perimeter
6     public static double checkTriangleAndGetPerimeter(double a, double b, double c) {
7         // Triangle validity conditions
8         if (a + b > c && a + c > b && b + c > a) {
9             // Valid triangle: calculate and return perimeter
10            double perimeter = a + b + c;
11            System.out.println("Valid triangle");
12            return perimeter;
13        } else {
14            // Invalid triangle
15            System.out.println("Invalid triangle");
16            return -1; // Return -1 to indicate invalid triangle
17        }
18    }
19
20    public static void main(String[] args) {
21        Scanner input = new Scanner(System.in);
22
23        System.out.print("Enter first side: ");
24        double side1 = input.nextDouble();
25
26        System.out.print("Enter second side: ");
27        double side2 = input.nextDouble();
28
29        System.out.print("Enter third side: ");
30        double side3 = input.nextDouble();
31
32        double perimeter = checkTriangleAndGetPerimeter(side1, side2, side3);
33    }
34 }

```

```

C:\WINDOWS\SYSTEM32\cmd
Enter first side: 23
Enter second side: 21
Enter third side: 20
Valid triangle
Perimeter: 64.00

-----
(program exited with code: 0)

Press any key to continue . . .

```

39.

The screenshot shows an IDE with a file named `Demo.java`. The code defines a `Demo` class with three overloaded `sum` methods and a `main` method for testing. The terminal window shows the output of the program.

```

1 public class Demo {
2
3     // Method for two integers
4     public static void sum(int a, int b) {
5         System.out.println("Sum of integers: " + (a + b));
6     }
7
8     // Overloaded method for two doubles
9     public static void sum(double a, double b) {
10        System.out.println("Sum of doubles: " + (a + b));
11    }
12
13    // Overloaded method for two Strings
14    public static void sum(String a, String b) {
15        System.out.println("Concatenated string: " + a + b);
16    }
17
18    // Main method for testing
19    public static void main(String[] args) {
20        sum(10, 20); // Calls int version
21        sum(5.5, 4.2); // Calls double version
22        sum("Hello", "World"); // Calls String version
23    }
24 }
25

```

```

C:\WINDOWS\SYSTEM32\cmd
Sum of integers: 30
Sum of doubles: 9.7
Concatenated string: HelloWorld

-----
(program exited with code: 0)
Press any key to continue . . . |

```

40.

The screenshot shows an IDE with a file named `Demo.java`. The code defines a `Demo` class with two overloaded `absoluteDifference` methods and a `main` method for testing. The terminal window shows the output of the program.

```

1 public class Demo {
2
3     // Method for two integers
4     public static int absoluteDifference(int a, int b) {
5         return Math.abs(a - b);
6     }
7
8     // Overloaded method for two doubles
9     public static double absoluteDifference(double a, double b) {
10        return Math.abs(a - b);
11    }
12
13    // Main method for testing
14    public static void main(String[] args) {
15        // Testing with integers
16        int intResult = absoluteDifference(10, 3);
17        System.out.println("Absolute difference (int): " + intResult);
18
19        // Testing with doubles
20        double doubleResult = absoluteDifference(5.5, 2.3);
21        System.out.println("Absolute difference (double): " + doubleResult);
22    }
23 }
24

```

```

C:\WINDOWS\SYSTEM32\cmd
Absolute difference (int): 7
Absolute difference (double): 3.2

-----
(program exited with code: 0)
Press any key to continue . . . |

```

41.

The screenshot shows an IDE with a file named `Demo.java`. The code defines methods for calculating the perimeter of a circle, rectangle, square, and triangle, and a `main` method that tests these methods. The output in the terminal window is as follows:

```

C:\WINDOWS\SYSTEM32\cmd
Circle perimeter (radius 5): 31.42
Rectangle perimeter (length 4, width 6): 20.0
Square perimeter (side 7): 28.0
Triangle perimeter (sides 3, 4, 5): 12.00

-----
(program exited with code: 0)
Press any key to continue . . . |

```

42.

```
public class PalindromeCheck {
```

```
    // Method for integer palindrome check
```

```
    public static boolean isPalindrome(int number) {
```

```
        int original = number;
```

```
        int reversed = 0;
```

```
        // Handle negative numbers (optional: treat as not palindrome)
```

```
        if (number < 0) {
```

```
            return false;
```

```
        }
```

```
        while (number != 0) {
```

```
            int digit = number % 10;
```

```
            reversed = reversed * 10 + digit;
```

```
            number /= 10;
```

```
        }
```

```
        return original == reversed;
```

```
    }
```

```

// Overloaded method for String palindrome check
public static boolean isPalindrome(String word) {
    if (word == null) {
        return false;
    }

    // Remove spaces and convert to lowercase for case-insensitive check
    String cleaned = word.replaceAll("\\s+", "").toLowerCase();

    int left = 0;
    int right = cleaned.length() - 1;

    while (left < right) {
        if (cleaned.charAt(left) != cleaned.charAt(right)) {
            return false;
        }
        left++;
        right--;
    }
    return true;
}

// Main method for testing
public static void main(String[] args) {
    // Testing integer palindrome
    System.out.println("121 is palindrome: " + isPalindrome(121)); // true
    System.out.println("123 is palindrome: " + isPalindrome(123)); // false
    System.out.println("1221 is palindrome: " + isPalindrome(1221)); // true
    System.out.println("-121 is palindrome: " + isPalindrome(-121)); // false

    // Testing String palindrome
    System.out.println("\"racecar\" is palindrome: " + isPalindrome("racecar")); // true
    System.out.println("\"hello\" is palindrome: " + isPalindrome("hello")); // false
    System.out.println("\"A man a plan a canal panama\" is palindrome: " +

```

```

        isPalindrome("A man a plan a canal panama"));          // true
    System.out.println("\n\"Madam\" is palindrome: " + isPalindrome("Madam"));    // true
}
}

```

43.

```

public class NumberConversion {

    // Decimal to Binary
    public static String convert(int decimal) {
        return Integer.toBinaryString(decimal);
    }

    // Overloaded: Decimal to Octal
    public static String convertToOctal(int decimal) {
        return Integer.toOctalString(decimal);
    }

    // Overloaded: Decimal to Hexadecimal
    public static String convertToHex(int decimal) {
        return Integer.toHexString(decimal).toUpperCase();
    }

    public static void main(String[] args) {
        int num = 25;
        System.out.println("Binary: " + convert(num));
        System.out.println("Octal: " + convertToOctal(num));
        System.out.println("Hexadecimal: " + convertToHex(num));
    }
}

```

44.

```
public class Factorial {  
    public static int factorial(int n) {  
        if (n < 0) return -1; // error case  
  
        int result = 1;  
  
        for (int i = 2; i <= n; i++) {  
            result *= i;  
        }  
  
        return result;  
    }  
  
    public static void main(String[] args) {  
        System.out.println(factorial(5)); // 120  
    }  
}
```

45.

```
public class ReverseString {  
    public static String reverse(String str) {  
        if (str == null || str.length() <= 1) {  
            return str;  
        }  
  
        return reverse(str.substring(1)) + str.charAt(0);  
    }  
  
    public static void main(String[] args) {  
        System.out.println(reverse("hello")); // olleh  
    }  
}
```

46.

```
public class GeometricSeries {  
  
    public static void printGeometric(int a, int r, int n, int current) {  
        if (current > n) return;  
        System.out.print(a * Math.pow(r, current - 1));  
        if (current < n) System.out.print(", ");  
        printGeometric(a, r, n, current + 1);  
    }  
  
    public static void main(String[] args) {  
        int a = 2, r = 3;  
        printGeometric(a, r, 10, 1);  
    }  
}
```

47.

```
public class PowerRecursion {  
  
    public static int power(int base, int exp) {  
        if (exp == 0) return 1;  
        return base * power(base, exp - 1);  
    }  
  
    public static void main(String[] args) {  
        System.out.println(power(2, 5)); // 32  
    }  
}
```

48.

```
public class SumOfDigits {  
  
    public static int sumDigits(int n) {  
        n = Math.abs(n);  
        if (n == 0) return 0;  
        return (n % 10) + sumDigits(n / 10);  
    }  
}
```

```
public static void main(String[] args) {  
    System.out.println(sumDigits(1234)); // 10  
    System.out.println(sumDigits(-123)); // 6  
}  
}
```

49.

```
import java.util.Scanner;
```

```
import java.util.Random;
```

```
public class GuessingGame {
```

```
    public static void guessNumber() {
```

```
        Random rand = new Random();
```

```
        Scanner input = new Scanner(System.in);
```

```
        int randomNum = rand.nextInt(101); // 0 to 100 inclusive
```

```
        int attempts = 5;
```

```
        boolean guessedCorrectly = false;
```

```
        System.out.println("I have generated a random number between 0 and 100.");
```

```
        System.out.println("You have 5 attempts to guess it.");
```

```
        for (int i = 1; i <= attempts; i++) {
```

```
            System.out.print("Attempt " + i + " - Enter your guess: ");
```

```
            int guess = input.nextInt();
```

```
            if (guess == randomNum) {
```

```
                System.out.println("Correct!");
```

```
                guessedCorrectly = true;
```

```
                break;
```

```
            } else if (guess > randomNum) {
```

```
                System.out.println("Too high, try again.");
```

```
    } else {  
        System.out.println("Too low, try again.");  
    }  
}
```

```
if (!guessedCorrectly) {  
    System.out.println("Sorry! You've used all 5 attempts.");  
    System.out.println("The correct number was: " + randomNum);  
}  
}
```

```
public static void main(String[] args) {  
    guessNumber();  
}  
}
```

50.

```
import java.util.Scanner;
```

```
public class Me {
```

```
    private static final String PASSWORD = "java123"; // password variable  
    private static int attemptCount = 0;  
    private static Scanner input = new Scanner(System.in);
```

```
// Method to check password
```

```
public static void checkPassword() {  
    attemptCount++;
```

```
    if (attemptCount > 3) {  
        System.out.println("Maximum attempts reached. Program ending.");  
        System.exit(0);  
    }
```

```

System.out.print("Enter password: ");

String enteredPassword = input.nextLine();

if (enteredPassword.equals(PASSWORD)) {
    System.out.println("Password correct!");
    displayDetails();
} else {
    System.out.println("Incorrect password.");
    if (attemptCount >= 3) {
        System.out.println("Maximum attempts reached. Program ending.");
        System.exit(0);
    }
    System.out.print("Do you want to continue (Yes/No)? ");
    String response = input.nextLine();
    if (response.equalsIgnoreCase("Yes")) {
        checkPassword(); // recursive call
    } else {
        System.out.println("Program ended.");
        System.exit(0);
    }
}
}

```

```

// Method to display user details
public static void displayDetails() {
    System.out.println("\n--- User Details ---");
    System.out.println("Name: John Doe");
    System.out.println("Email: john.doe@example.com");
    System.out.println("Role: Student");
    System.out.println("Program ended successfully.");
    System.exit(0);
}

```

```

public static void main(String[] args) {

```

```
checkPassword(); // first invocation
```

```
}
```

```
}
```